

PREHEATER CYCLONE PROCESS INTELLIGENCE

Abnormal Operating Period Detection from Historical Sensor Data



About Data Set:

Dataset	Details
Observations	377,719
Process variables	6
Sampling	Every 5 minutes
Period	January 2017 – August 2020
Approach	Feature engineering + Isolation Forest + persistence filtering

Purpose: identify sustained periods where overall process behaviour is unusual, rather than treating every unusual sensor reading as a fault.

Objective: - Identify time periods where the cyclonepreheater operates differently from its normal processbehaviour using historical sensor data

1. Executive Summary

This project develops an unsupervised method to identify abnormal operating periods in a cyclone preheater using historical process sensor data. The dataset contains 377,719 observations recorded at five-minute intervals between January 2017 and August 2020 across six temperature and draft variables.

Because the dataset has no labelled normal/abnormal events, the solution does not predict a predefined fault class. Instead, it learns unusual multivariate process behaviour and highlights time periods that deserve engineering review.

The workflow combines data-quality checks, process-oriented feature engineering and Isolation Forest. The final model uses 36 features representing current sensor values, sensor relationships, short/medium-term changes and local variability. A 99th-percentile anomaly threshold identifies candidate observations, and a 30-minute persistence rule converts isolated points into sustained abnormal periods.

Key metric	Result
Modelled observations	374,761 (99.22% of dataset)
Final model features	36
Isolation Forest trees	200
Threshold	99th percentile ≈ 0.5977
Candidate anomaly observations	3,748
Persistence requirement	30 minutes
Final abnormal periods	252
Total detected duration	8 days 3 hours 55 minutes
Highest anomaly score	0.7406

2. Problem Statement and Objective

The assignment is to use Python and an appropriate algorithm to highlight time periods where abnormality can be observed in cyclone preheater sensor data. The desired output is therefore time-based: start time, end time, duration and severity.

Objectives:

- Detect unusual behaviour across multiple process sensors.
- Capture both current operating condition and recent process dynamics.
- Avoid treating every extreme value as abnormal.
- Convert individual anomaly observations into sustained periods.
- Produce a screening list that engineers can investigate further.

The result is described as candidate abnormal operating periods, not confirmed equipment failures.

3. Dataset and Process Variables

Variable	Group	Role
Cyclone_Inlet_Gas_Temp	Temperature	Inlet gas thermal condition
Cyclone_Gas_Outlet_Temp	Temperature	Gas outlet thermal condition
Cyclone_Material_Temp	Temperature	Material thermal condition
Cyclone_Outlet_Gas_draft	Draft	Outlet gas draft behaviour
Cyclone_cone_draft	Draft	Cone draft behaviour
Cyclone_Inlet_Draft	Draft	Inlet draft behaviour

The sensors are strongly related. The two gas-temperature variables correlate at about 0.99, while Outlet Gas Draft and Inlet Draft are approximately 1.00 correlated in the plotted matrix. This supports treating the system as a multivariate process rather than six independent signals.

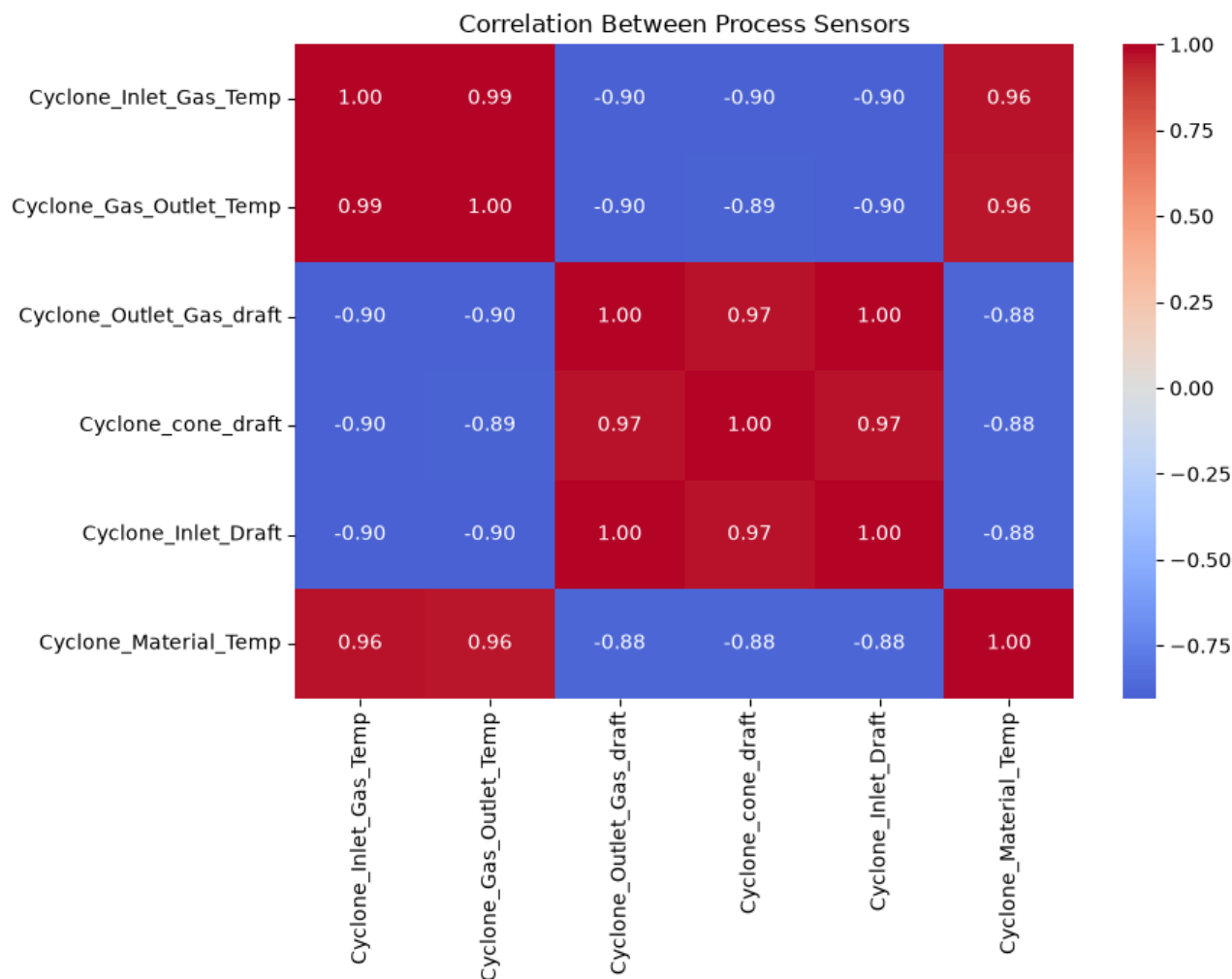


Figure 1. Correlation between the six process sensors.

4. Overall Project Workflow

The project follows a simple pipeline: clean the signals → understand normal process relationships → describe recent behaviour → detect unusual combinations → require persistence → report abnormal periods.

Stage	What happens	Main output
1. Raw data	Load timestamped sensor records	377,719 observations
2. Preparation	Parse time, convert sensors, separate status strings	Clean numeric sensor data
3. Quality checks	Check missing values, duplicates, gaps and unusual values	Validated input
4. Process understanding	Study distributions, correlations and regimes	Context for modelling
5. Feature engineering	Create relationships, changes and rolling variability	44-column feature dataset
6. Model	Train 200-tree Isolation Forest	Point anomaly scores
7. Threshold	Use 99th-percentile score	3,748 candidate points
8. Persistence	Group consecutive 5-minute anomalies; require 6 points	Sustained periods
9. Final output	Calculate start, end, duration and severity	252 abnormal periods

10. Engineering review	Inspect strongest periods and process charts	Investigation candidates
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5. Data Preparation and Quality Checks

5.1 Timestamp and continuity

The raw time column contained mixed timestamp formats. Mixed-format parsing standardized the timestamps successfully, with no invalid timestamps remaining. The data runs from 1 January 2017 to 7 August 2020.

- 377,715 intervals were exactly five minutes apart.
- There were no duplicate timestamps.
- Only three one-day gaps were identified.

5.2 Sensor status values

Raw sensor columns also contain system/status strings such as Not Connect, I/O Timeout, Configure, Scan Timeout, Comm Fail and Unit Down. These were separated from numeric process readings instead of being assigned artificial numeric values.

5.3 Missing numeric values

Sensor	Missing	Rate
Inlet Gas Temp	1,320	0.349%
Material Temp	1,591	0.421%
Outlet Gas draft	1,321	0.350%
Cone draft	1,320	0.349%
Gas Outlet Temp	1,321	0.350%
Inlet Draft	1,322	0.350%

Rows with incomplete model features are excluded from model fitting rather than filling sensor values arbitrarily. This leaves 374,761 usable observations.

5.4 Extreme values and context

Unusual values were not automatically deleted. For example, negative draft values are common, and Material Temperature values around 0 or 900 can occur with coherent readings from other sensors. This supports a context-aware multivariate method rather than simple fixed thresholds.

6. Exploratory Data Analysis and Process Understanding

6.1 Temperature trends

The long-range temperature plot shows recurring operating regimes and transitions. Therefore, a low temperature value alone cannot be considered abnormal; the surrounding process state matters.

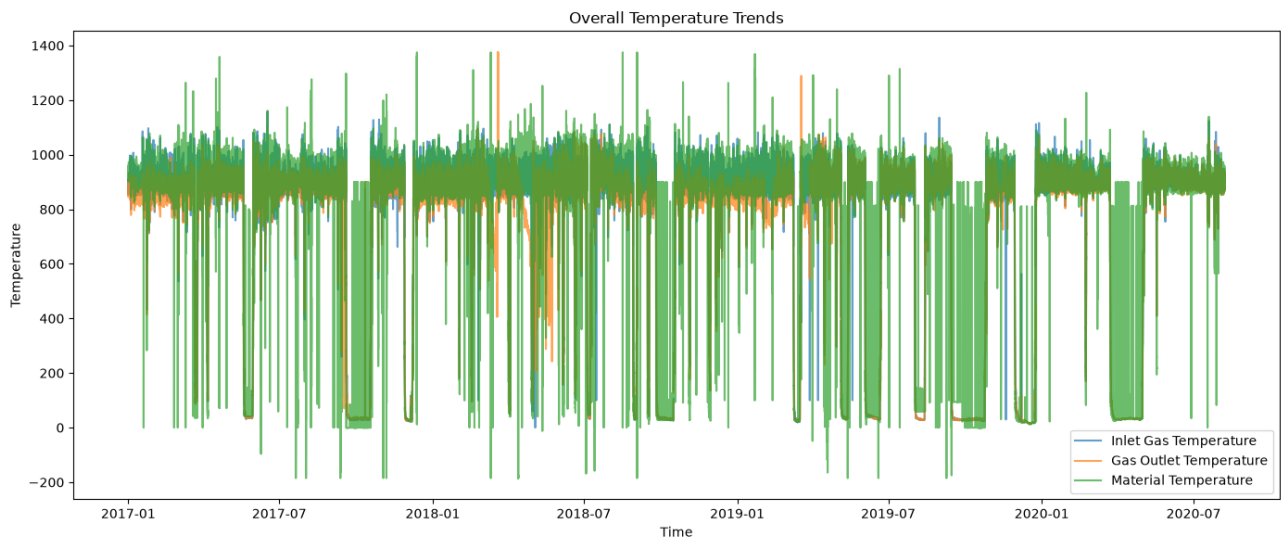


Figure 2. Overall temperature trends.

6.2 Draft trends

The draft signals also show recurring states. Most values are strongly negative, but the process periodically moves toward zero or positive values. These coordinated transitions are important for anomaly detection.

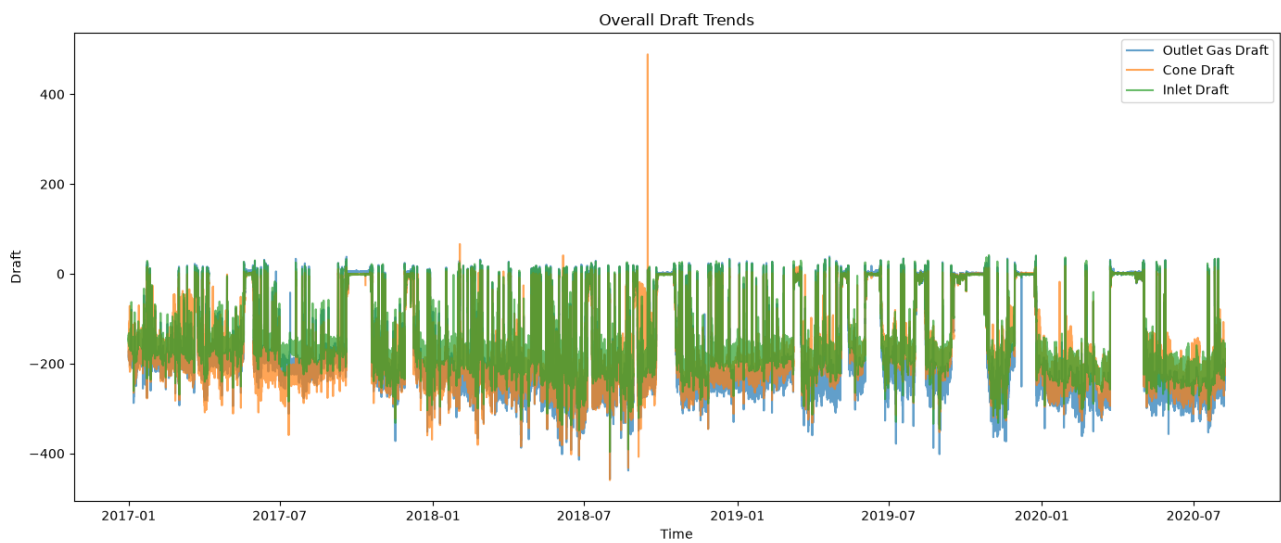


Figure 3. Overall draft trends.

6.3 Draft sensor relationship

The Outlet Gas Draft versus Inlet Draft scatter plot shows a very strong approximately linear relationship, supporting relationship-based features.

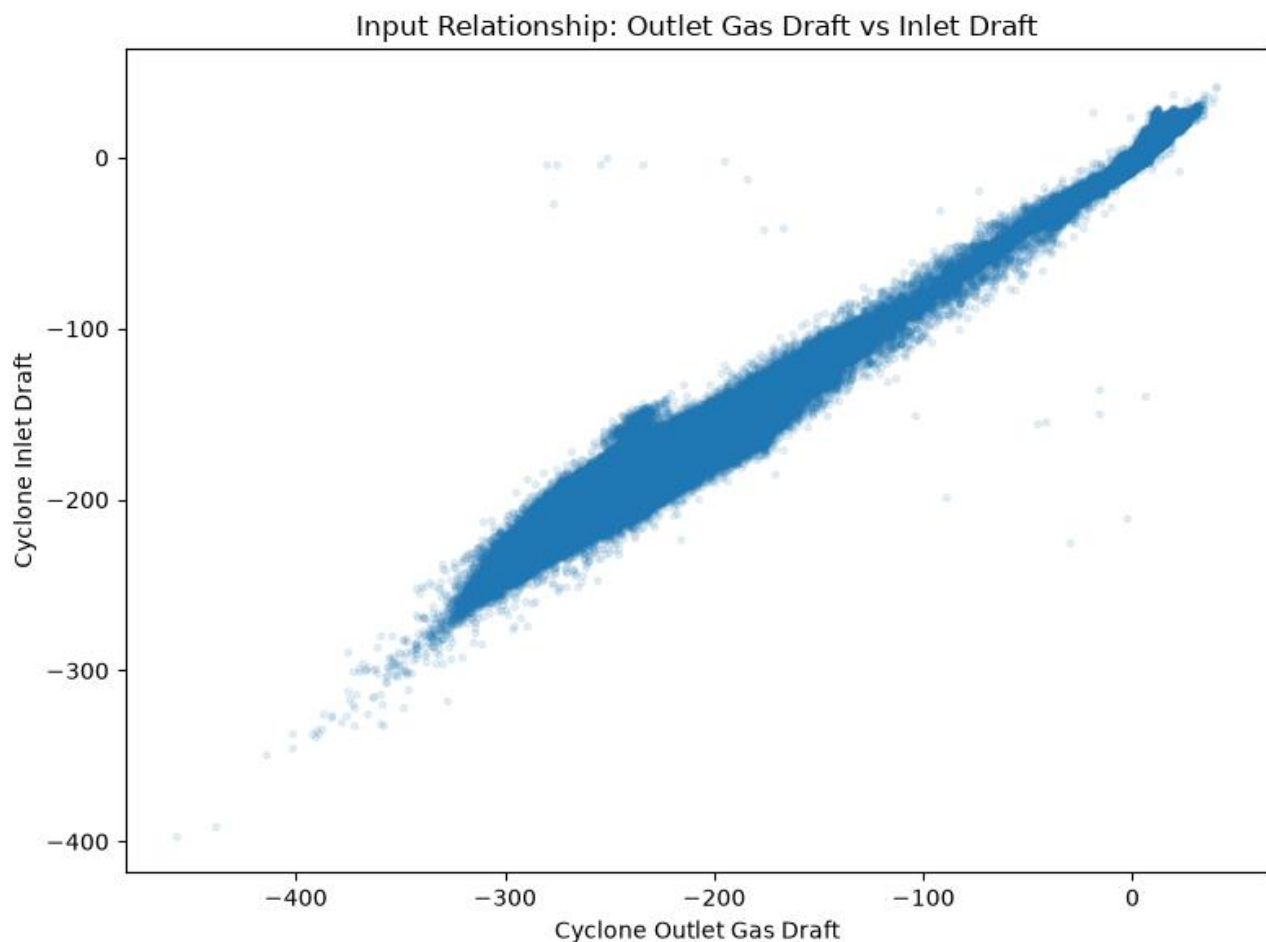


Figure 4. Outlet Gas Draft versus Inlet Draft.

7. Feature Engineering

Raw sensor values describe where the process is now. Feature engineering adds information about how the process is changing and how variable it has been recently.

7.1 Relationship features

Six relationship features were created to capture interactions between temperature and draft measurements. These help the model recognize unusual combinations even when individual readings are not extreme.

7.2 Change features

For every sensor, 5-, 30- and 60-minute changes were calculated, producing 18 dynamic features. The short window captures rapid movement, while the longer windows capture sustained transitions.

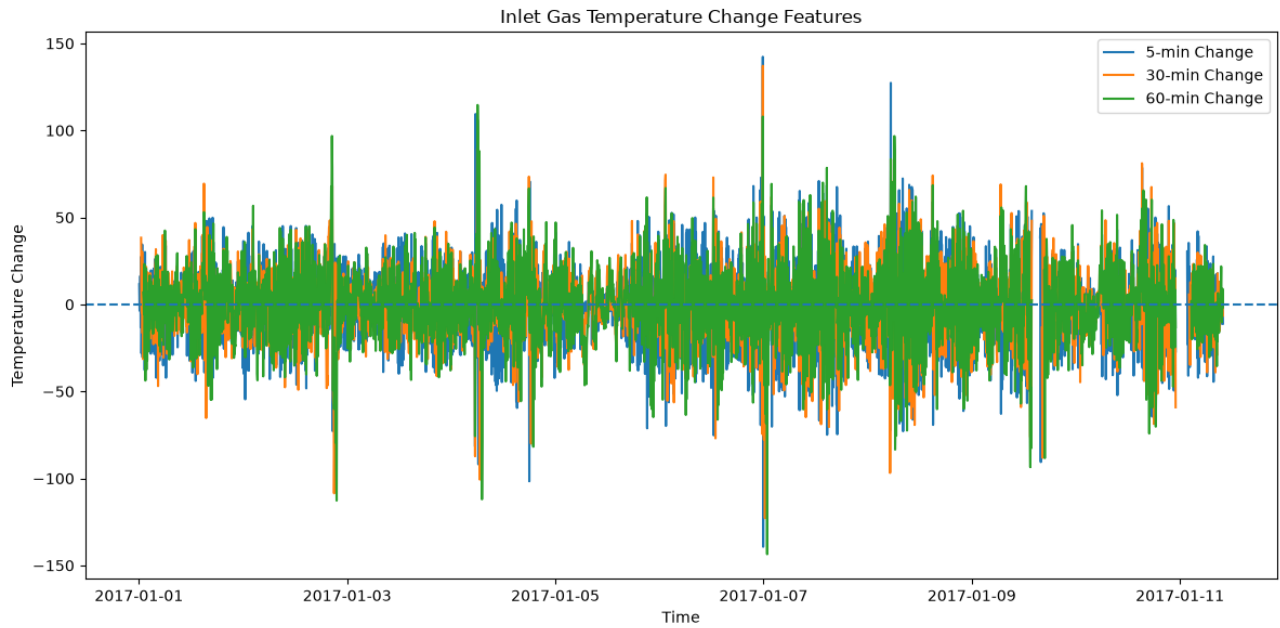


Figure 5. Example of 5-, 30- and 60-minute temperature change features.

7.3 Rolling statistics

A 60-minute rolling mean and rolling standard deviation were created for every sensor. The rolling mean represents local process level; rolling standard deviation represents local fluctuation.

The rolling mean was later excluded from the final Isolation Forest inputs because it was highly redundant with the raw sensor values. Rolling standard deviation was retained because it adds local variability information.

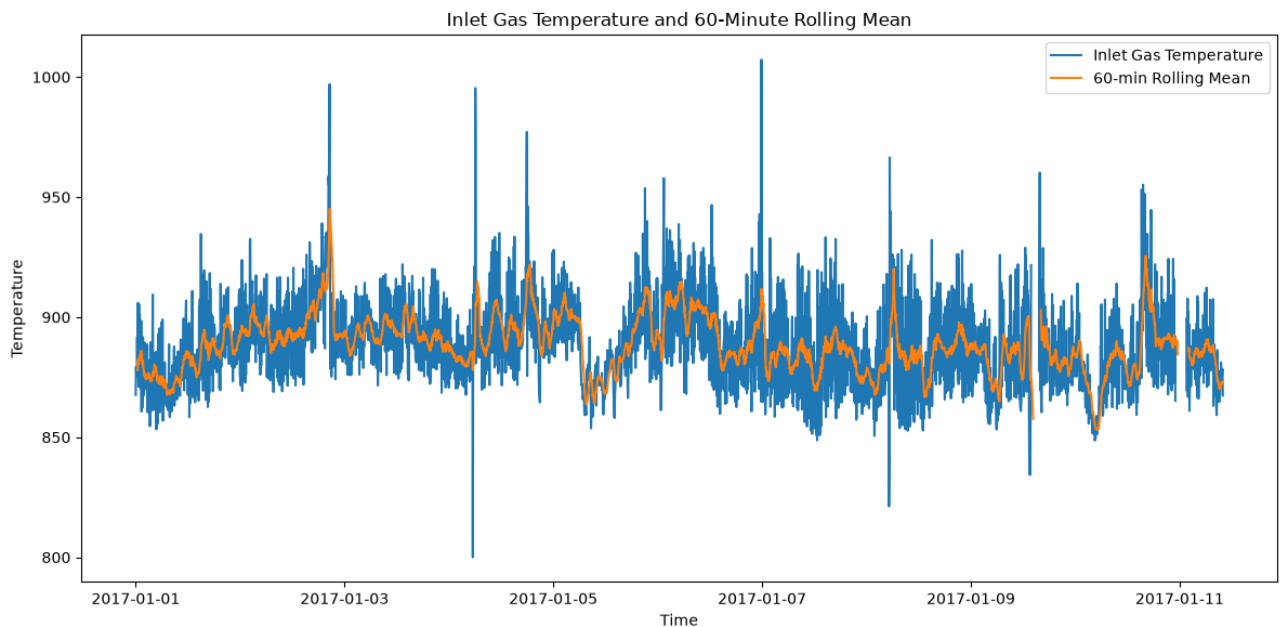


Figure 6. Inlet Gas Temperature with its 60-minute rolling mean.

7.4 Final model inputs

The feature-engineered dataset contains 44 columns overall. The final model uses 36 features: 6 raw sensor values, 6 relationship features, 18 change features and 6 rolling-standard-deviation features. The temperature-based operating regime was created for context but was not included in the final 36 model inputs.

8. Anomaly Detection Method

8.1 Why Isolation Forest?

Isolation Forest was selected because there are no labelled abnormal events. It is an unsupervised method designed to isolate observations that are unusual relative to the rest of the dataset.

- Random Forest is primarily supervised and would require labelled outcomes.
- K-Means is useful for clustering operating states but is not directly an anomaly-ranking method.
- An Autoencoder could model nonlinear behaviour but would add unnecessary neural-network complexity for this assignment.
- Isolation Forest gives a simple anomaly-score workflow that is easy to explain and reproduce.

8.2 Configuration

Setting	Choice
Algorithm	Isolation Forest
Trees	200
Contamination	auto
Random state	42
Model inputs	36 features
Score	Higher = more unusual
Threshold	99th percentile ≈ 0.5977

8.3 Interpretation

The model does not identify a specific physical fault. It identifies observations whose multivariate feature pattern is unusual. Because the features include changes and rolling variability, the score reflects both unusual values and unusual recent behaviour.

9. From Anomaly Points to Abnormal Periods

The 99th-percentile threshold produced 3,748 candidate anomaly observations out of 374,761 modelled observations. An individual point can be noise or a transient measurement, so it is not directly reported as an abnormal period.

Consecutive anomaly observations were grouped only when timestamps remained five minutes apart. A group had to contain at least six observations, equal to 30 minutes, to become a final abnormal period. This converts point-level detection into a process-oriented time-period output.

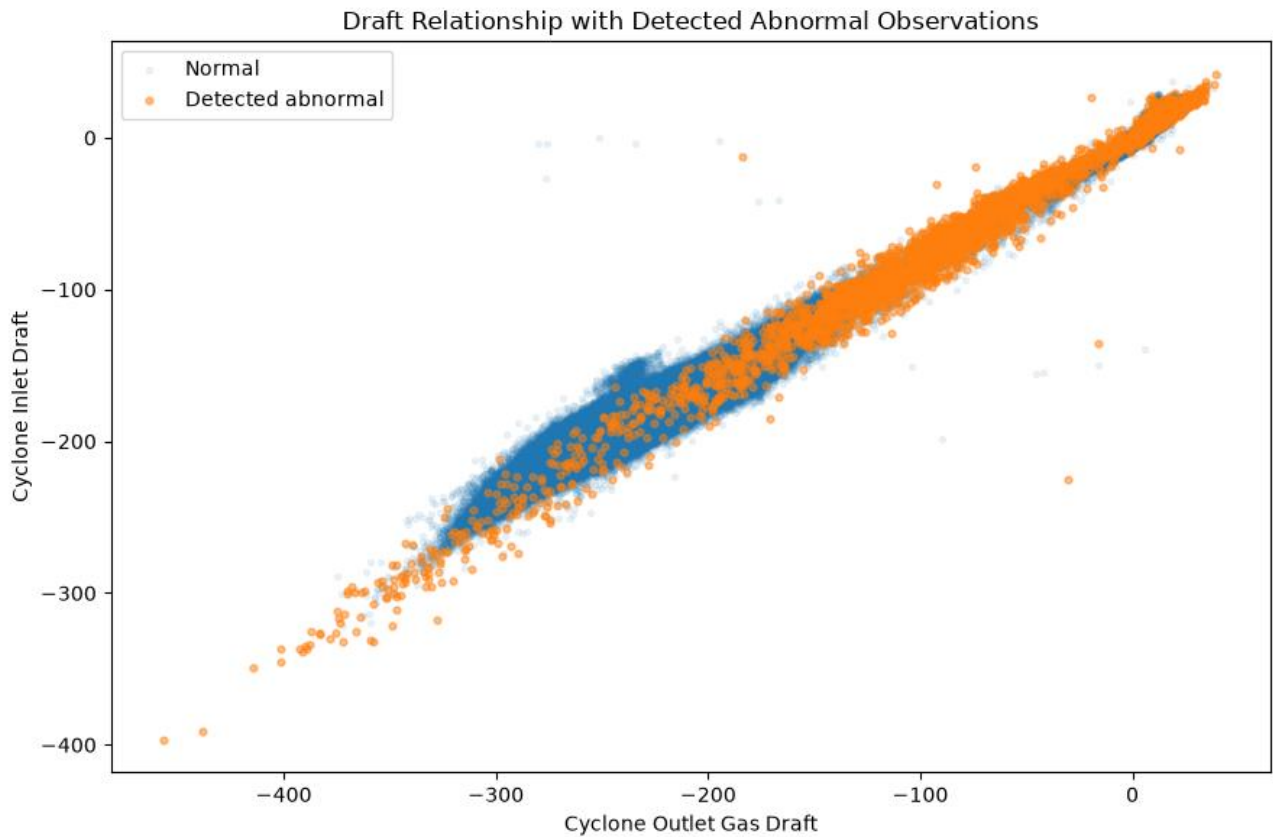


Figure 7. Draft relationship with detected abnormal observations highlighted.

10. Results and Findings

Result	Value
Candidate anomaly observations	3,748
Minimum duration	30 minutes
Final abnormal periods	252
Total detected duration	8 days 3 hours 55 minutes
Highest mean score	0.7030
Highest score	0.7406

10.1 Results by year

Year	Periods	Total duration	Highest mean score
2017	80	2 days 14 hours 30 minutes	0.6811
2018	102	3 days 7 hours 10 minutes	0.7030
2019	49	1 day 12 hours 35 minutes	0.6804
2020	21	17 hours 40 minutes	0.6785

2018 contains the highest number of detected periods and the highest total detected duration. This should not be interpreted as proof that 2018 was the worst operating year; it is simply the result of the selected detection rule.

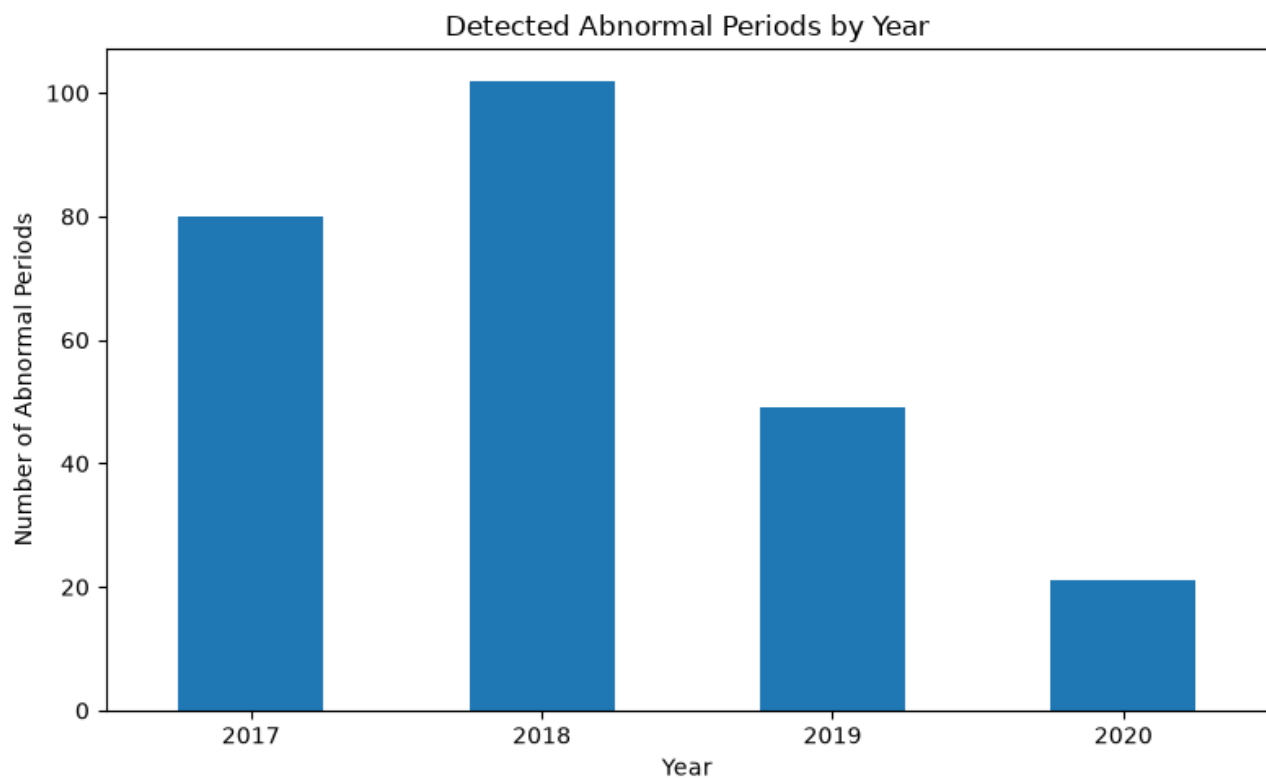


Figure 8. Detected abnormal periods by year.

10.2 Strongest detected periods

The top periods are ranked by mean anomaly score, so the ranking reflects how unusual the process remained throughout each detected period.

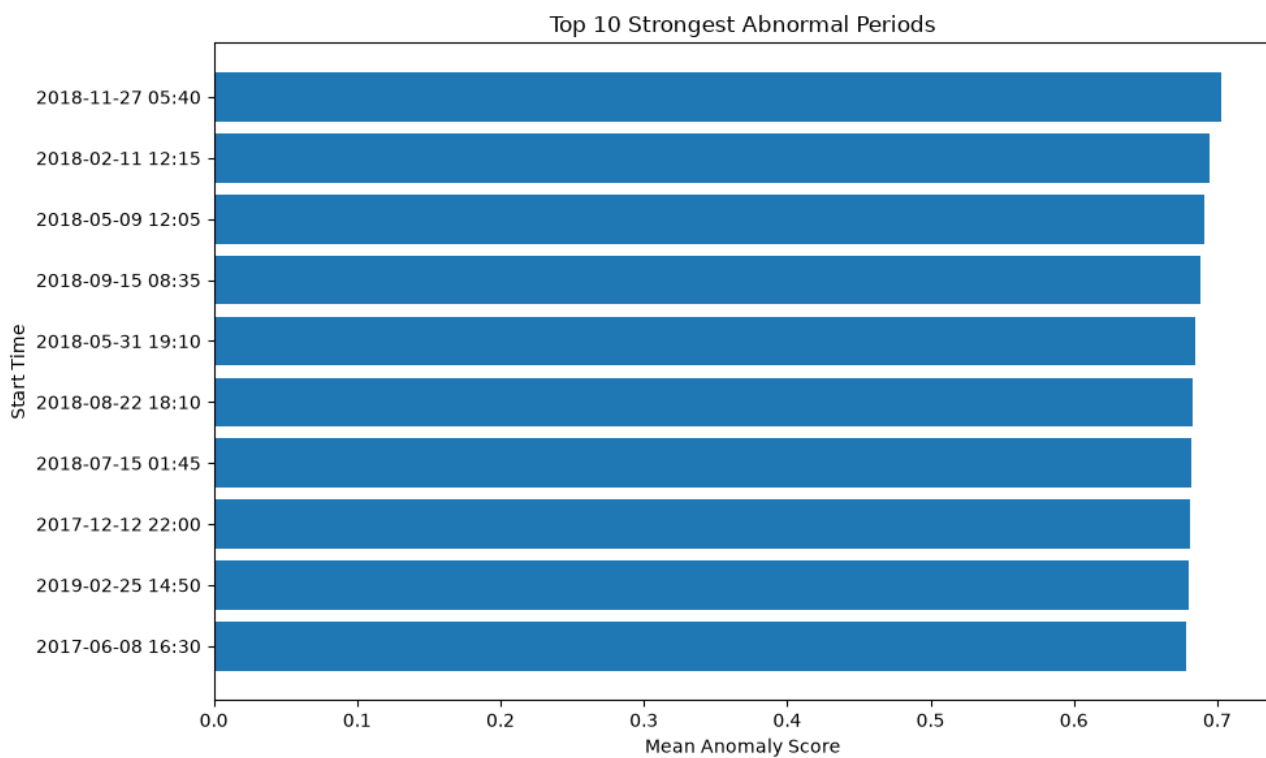


Figure 9. Top 10 strongest abnormal periods by mean anomaly score.

11. Strongest Detected Abnormal Period

The strongest period occurs on 27 November 2018 from 05:40 to 06:30. It lasts 50 minutes, with a mean anomaly score of 0.7030 and a maximum of 0.7406.

Variable	1 hour before	During period	Observed change
Inlet Gas Temperature	911.28	750.47	Large decrease
Gas Outlet Temperature	866.23	716.14	Large decrease
Material Temperature	951.90	547.53	Large decrease
Outlet Gas Draft	-252.02	19.52	Shift toward zero/positive
Cone Draft	-252.01	-4.23	Shift toward zero
Inlet Draft	-193.86	16.99	Shift toward zero/positive

These comparisons show that several variables moved together. The temperature group fell substantially while the draft group shifted from strongly negative values toward zero or positive values.

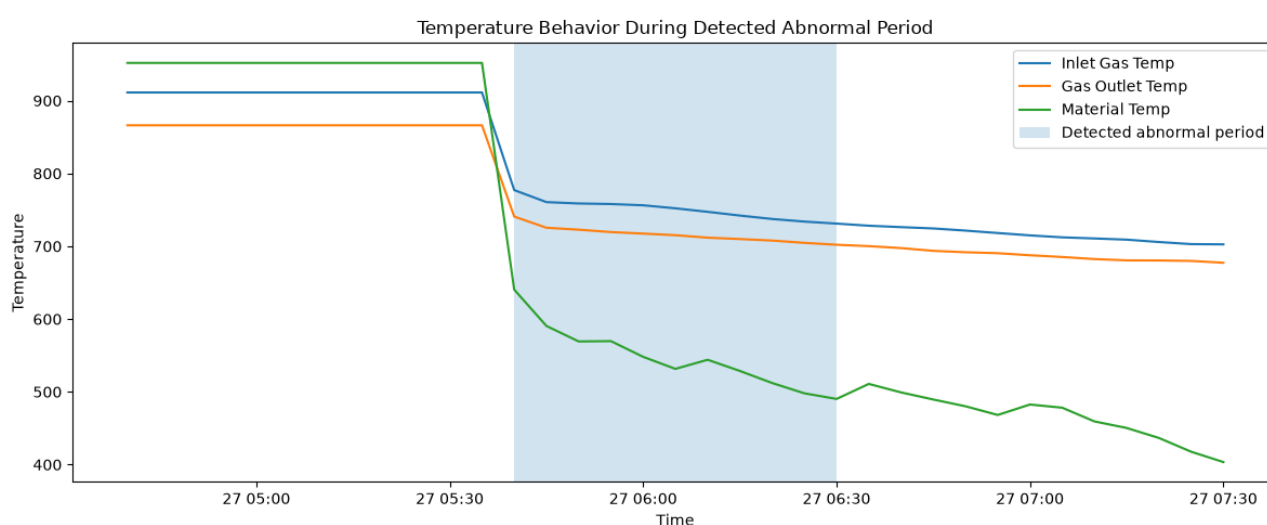


Figure 10. Temperature behaviour during the strongest detected abnormal period.

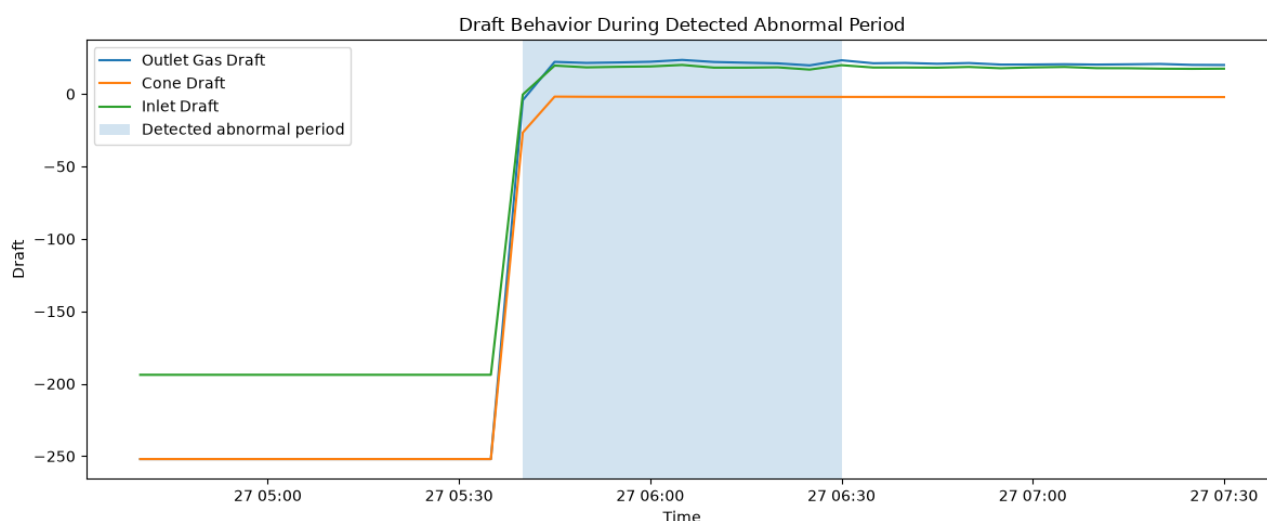


Figure 11. Draft behaviour during the strongest detected abnormal period.

The shaded region is the detected 05:40–06:30 period. The plots show a coordinated multivariate transition rather than a single isolated sensor spike. This is a strong candidate for further process investigation, but the analysis does not by itself establish the physical root cause.

12. Interpretation of the Visual Results

Overall trends

The temperature and draft plots show multiple recurring operating regimes. This is why simple absolute thresholds would be risky: the same value can be normal in one operating state and unusual in another.

Change features

The 5-, 30- and 60-minute changes show both normal fluctuations and larger transitions. Longer windows help distinguish sustained movement from one five-minute spike.

Rolling statistics

The rolling mean follows the local process baseline and smooths short-term noise. Rolling standard deviation captures local variability and was retained in the model.

Draft relationships

The strong draft relationship confirms that the draft sensors share a common process signal. Detected anomalies can therefore represent coordinated shifts across the draft group.

13. Limitations and Engineering Considerations

- The model is unsupervised, so detected periods are candidate abnormal operating conditions rather than confirmed faults.
- The 99th-percentile threshold is a practical screening choice and may be changed for a production use case.
- The 30-minute persistence rule is a modelling choice and should be aligned with actual process response times.
- Highly correlated sensors provide overlapping information; redundant rolling means were therefore excluded from the final model.
- The detector identifies unusual behaviour but cannot diagnose the physical root cause without process logs, maintenance records or engineering validation.
- The industrial dataset should only be placed in a public repository if its owner permits disclosure.

14. Reproducibility and Project Structure

Notebook / output	Purpose
1_data_preprocessing.ipynb	Timestamp parsing, numeric conversion, status separation and quality checks
2_feature_engineering.ipynb	Relationship, change and rolling-statistic features
3_anomaly_detection.ipynb	Isolation Forest, scores, threshold and period grouping
4_visualizations.ipynb	EDA, feature and model-result visualizations
cleaned_sensor_data.csv	Cleaned sensor data
feature_engineered_data.csv	Feature-engineered dataset
detected_abnormal_periods.csv	Final 252 abnormal periods
anomaly_scores.csv	Optional point-level anomaly scores

Reproduction sequence: run preprocessing → feature engineering → anomaly detection → visualizations.

The main operational output is detected_abnormal_periods.csv. Each row represents a sustained candidate period with start time, end time, duration and anomaly severity information.

15. Conclusion

This project combines process relationships, short- and medium-term changes, local variability and an unsupervised Isolation Forest model to detect sustained unusual behaviour in cyclone preheater data.

The final analysis identifies 252 sustained abnormal periods. The strongest event, on 27 November 2018, shows a coordinated shift in temperature and draft behaviour. The resulting list is best viewed as an engineering screening tool: it narrows a large historical dataset into specific time periods that can be investigated against plant logs, operating events and maintenance records.

Key takeaway: detect sustained unusual process behaviour first, then use engineering context to determine why it happened.